

Optimization

Minimization and Karnaugh Maps

- Simplify (or minimize) POS functions (circuits):

Example 1: Find the minimum cost POS expression for the following using a K-map

$$f(x_1, x_2, x_3) = \overline{x_1} \overline{x_2} x_3 + \overline{x_1} x_2 x_3 + x_1 \overline{x_3} \rightarrow \text{SOP}$$

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$x_1 x_2 \backslash x_3$	00	01	11	10
0	0 ⁰	0 ²	1 ⁶	1 ⁴
1	1 ¹	0 ³	0 ⁷	1 ⁵

$$(x_1 + x_3)$$

$$(\overline{x_2} + \overline{x_3})$$

$$f = (x_1 + x_3) \cdot (\overline{x_2} + \overline{x_3}) \text{ POS}$$

Example 2: Give the minimum cost POS representation of the following logical expression

$$\underline{f(x_1, x_2, x_3) = \prod M(0, 1, 4, 5, 6)} \rightarrow \text{pos}$$

x_1, x_2	00	01	11	10
x_3				
0	0 ⁰	1 ²	0 ⁶	0 ⁴
1	0 ¹	1 ³	1 ⁷	0 ⁵

Handwritten annotations: A blue circle around the 0s in the top row (00, 01, 11, 10) is labeled $(\bar{x}_1 + x_3)$. A red circle around the 0s in the bottom row (00, 01, 11, 10) is labeled x_2 .

$$f = x_2 \cdot (\bar{x}_1 + x_3)$$

Example 3: Find the minimum cost POS expression for the following function.

$$f(x_1, x_2, x_3, x_4) = \sum m(2, 3, 9, 10, 11, 13) \rightarrow \underline{\text{SOP}}$$

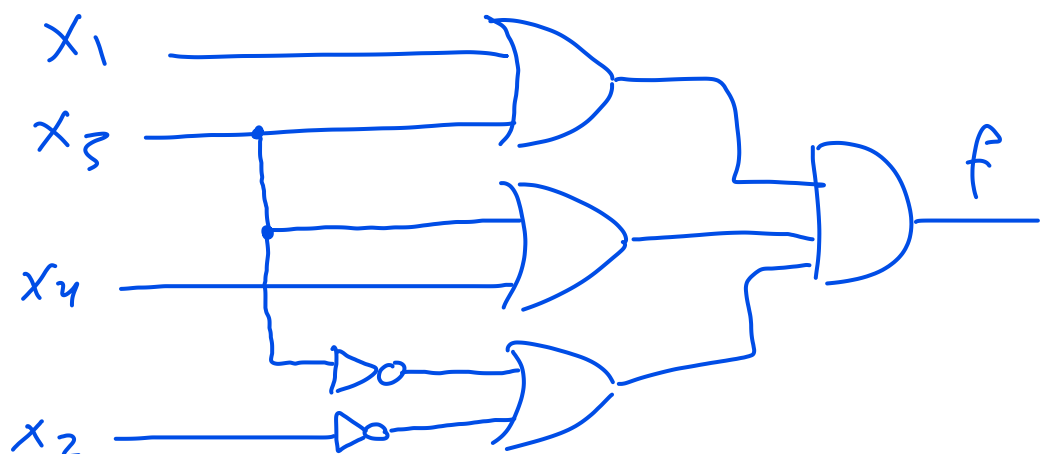
Handwritten annotations on the Karnaugh map:

- Red arrow from $(x_1 + x_3)$ points to the first column (00).
- Red arrow from $(x_3 + x_4)$ points to the top row (00).
- Red arrow from $(\bar{x}_2 + \bar{x}_3)$ points to the bottom row (10).

x_1, x_2 x_3, x_4	00	01	11	10
00	0 0	0 4	0 12	0 8
01	0 1	0 5	1 13	1 9
11	1 3	0 7	0 15	1 11
10	1 2	0 6	0 14	1 10

$$f = (x_1 + x_3) \cdot (x_3 + x_4) \cdot (\bar{x}_2 + \bar{x}_3)$$

Design (draw) the circuit



Don't Cares in K-Maps

- Any function may have input values that will never occur $\Rightarrow$ the output value is considered as a "don't care", X or d
- Don't cares in the K-map may reduce the cost of the logic circuit

Given $f(x_1, \dots, x_4) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$, find the minimum cost SOP and POS expressions.

$x_1x_2 \backslash x_3x_4$	00	01	11	10
00	0 ⁰	1 ⁴	d ¹²	0 ⁸
01	0 ¹	1 ⁵	d ¹³	0 ⁹
11	0 ³	0 ⁷	d ¹⁵	0 ¹¹
10	1 ²	1 ⁶	d ¹⁴	1 ¹⁰

$$f = x_3 \bar{x}_4 + x_2 \bar{x}_3 \rightarrow \text{min cost SOP}$$

$x_1x_2 \backslash x_3x_4$	00	01	11	10
00	0	1	d	0
01	0	1	d	0
11	0	0	d	0
10	1	1	d	1

$$f = (\bar{x}_3 + \bar{x}_4) \cdot (x_2 + x_3) \rightarrow \text{min cost POS}$$

Terminology

Literal: Each appearance of a variable, either uncomplemented or complemented, is called a literal.

$$a\bar{b} \rightarrow 2 \text{ literals}$$

$$a\bar{b} + ab\bar{c} \rightarrow 5 \text{ literals}$$

Implicant: A product term for which the function is equal to 1.

Example: How many potential implicants are there in this function, $f(x_1, x_2, x_3) = \sum m(3, 4, 5, 6, 7)$?

$$\textcircled{1} \quad \textcircled{11} \quad \textcircled{1} \quad \textcircled{11}$$

$$5 + 3 + 2 + 1$$

= 11 implicants

x_1, x_2		00	01	11	10
x_3	0	0	0	1	1
	1	0	1	1	1

Prime Implicant: An implicant that cannot be combined with another to reduce the number of literals and still be an implicant.

2 prime implicants

$$x_1, x_2x_3$$

x_1, x_2		00	01	11	10
x_3	0	0	0	1	1
	1	0	1	1	1

Red boxes highlight the prime implicants: x_2x_3 (covering cells (0,1), (0,0), (1,1), (1,0)) and x_1 (covering cells (1,1), (1,0), (1,1), (1,0)).

Essential Prime Implicant: If a prime implicant includes a minterm for which $f = 1$ that is not included in any other prime implicant, is called essential prime implicant.

2 essential prime implicants

$$x_1, x_2 x_3$$

$x_1 x_2$	00	01	11	10
x_3				
0	0	0	1	1
1	0	1	1	1

Cover: A collection of implicants that covers all the minterms of the function. A cover defines a particular implementation of the function. $f = \sum m(3, 4, 5, 6, 7) \rightarrow \text{Cover 1}$

$$f(x_1, x_2, x_3) = \bar{x}_1 x_2 x_3 + \underline{x_1 \bar{x}_2 \bar{x}_3} + \underline{x_1 \bar{x}_2 x_3} + x_1 x_2 \bar{x}_3 + x_1 x_2 x_3$$

$$= \underline{\bar{x}_1 x_2 x_3} + x_1 \bar{x}_2 + x_1 x_2 \bar{x}_3 + \underline{x_1 x_2 x_3} \rightarrow \text{Cover 2}$$

$$= x_2 x_3 + x_1 \bar{x}_2 + x_1 x_2 \bar{x}_3 \rightarrow \text{Cover 3}$$

$$f = x_2 x_3 + x_1 \rightarrow \text{Cover 4}$$

$x_1 x_2$	00	01	11	10
x_3				
0	0	0	1	1
1	0	1	1	1

$x_2 x_3$ x_1

Example:

The min cost cover

Given the following function:

Four-variable function $f(x_1, \dots, x_4) = \sum m(2, 3, 5, 6, 7, 10, 11, 13, 14)$.

1. How many possible implicants does the given function f have? **23**
2. Find all prime implicants. **5**
3. Find all essential prime implicants. **3 (red groups)**
4. Derive the minimum cost cover.

$$f = x_2 \bar{x}_3 x_4 + \bar{x}_2 x_3 + x_3 \bar{x}_4 + \bar{x}_1 x_3$$

